Site and Event Characterization

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Sites were selected to represent a diverse range of conditions relevant to $PM_{2.5}$ sensor network performance during smoke events, including:

- **Urban vs. rural settings** — Dense metropolitan areas to sparsely populated mountain valleys
- **Wildfire (WF) and prescribed fire (Rx) smoke** — 8 smoke events from wildfire sources and 3 smoke events from prescribed fire sources
- **Localized versus regional smoke influences** — Smoke transport from local smoke sources, as well as regional transport from distant wildfires
- **Topographically complex versus flat terrain** — Mountainous regions with deep drainages as well as low-relief plains
- **Dense versus sparse sensor networks** — from 719 devices (Los Angeles) to 18 devices (Duluth), with spatial connectivity ranging from a single inter-connected network of sensors, to fragmented configurations
- **Differing environmental and geographic regions** — Pacific Northwest, Northern Rockies, Great Basin steppe, Great Lakes, Canadian Prairie, Southern California coastal urban, and Sierra Nevada montane forest.

Each site is described below with terrain and land cover characteristics, sensor network composition and spatial density, and smoke/fire attribution during the event period. Summary maps show sensor locations by type, cumulative HMS satellite fire detections within 50 km, and the zone boundary. Timeline charts show daily fire detection counts and HMS smoke plume coverage.

> **NOTE REGARDING SENSOR/MONITOR COUNTS:** Device counts and spatial metrics reported here reflect the total inventory of unique sensors present in the training dataset for each zone. The presence of a sensor in this inventory does not guarantee that it was actively reporting during the event period. Sensors may have been offline, relocated, renamed, or replaced over time. The counts characterize the *potential* spatial coverage of each zone's sensor infrastructure, not the number of sensors that contributed data to any specific event analysis.

Table 1: Event Site Geographic Overview

Site Name	State/Province	Area (km²)	Elev. Mean (m)	Elev. Range (m)	Mean Slope (°)	Dominant NLCD Land Cover	Developed (%)	Forest (%)	Setting
Boise	Idaho	4,046	998	1,634	8.2	Grassland/Herbaceous	16.0	10.9	Suburban/Exurban
Calgary	Alberta	2,321	1,103	472	2.9	— (Canadian, no NLCD)	0.0	0.0	Urban/Suburban
Deschutes	Oregon	4,618	1,357	2,422	4.9	Evergreen Forest	6.3	47.7	Rural/Exurban
Duluth	Minnesota	1,860	305	285	2.2	Woody Wetlands	12.8	32.5	Small City/Rural
Helena	Montana	378	1,243	693	5.3	Grassland/Herbaceous	21.1	11.2	Small City/Rural
Los Angeles	California	7,960	173	1,882	5.6	Developed Medium Intensity	61.3	1.3	Dense Urban
Methow Valley	Washington	2,781	1,071	2,475	18.2	Grassland/Herbaceous	4.8	29.5	Rural/Wilderness
Missoula	Montana	1,003	1,269	1,225	15.7	Evergreen Forest	11.1	43.8	Small City/Rural
Spokane	Washington	971	654	653	5.7	Shrub/Scrub	39.6	20.2	Urban/Suburban
Truckee	California	260	1,954	876	9.5	Evergreen Forest	16.6	62.5	Rural/Wilderness

Land cover from NLCD 2021 (U.S. sites) via Google Earth Engine. Calgary land cover not available from NLCD. Elevation and slope derived from USGS 3DEP 30 m DEM (U.S.) and SRTM (Canada).

Table 2: Sensor Network Composition

Site	Total Devices	PurpleAir	Permanent Monitor	Temporary Monitor	SensOR	SensWA	Density (per 1,000 km²)
Boise	72	65	4	3	—	—	17.8
Calgary	40	36	4	—	—	—	17.2
Deschutes	258	241	4	8	5	—	55.8
Duluth	18	15	3	—	—	—	9.7
Helena	25	22	1	2	—	—	66.0
Los Angeles	719	712	7	—	—	—	90.4
Methow Valley	77	70	5	—	—	2	27.6
Missoula	58	42	1	15	—	—	57.7
Spokane	74	59	8	—	—	7	76.0
Truckee	99	95	2	2	—	—	379.9

Counts reflect all unique sensors in the training dataset; not all sensors were necessarily active during each event (see note above).

Table 3: Spatial Connectivity

Spatial connectivity metrics characterize how sensors are distributed within each zone. At a given radius threshold, two sensors that fall within that distance of each other are considered "connected." Connected components are groups of sensors linked transitively at that threshold — i.e., sensor A and sensor C belong to the same component if A is within range of B, and B is within range of C, even if A and C are not directly within range of each other. A single connected component means all sensors can reach each other through such chains; many components indicate networks clustered away from each other, such as in two separate population centers. Isolated sensors have zero neighbors within the given radius.

Site	Total Devices	Mean Neighbors (5 km)	Components (5 km)	Isolated (5 km)	Mean Neighbors (7.5 km)	Components (7.5 km)	Mean Neighbors (10 km)	Components (10 km)
Los Angeles	719	23.4	15	4	44.2	2	69.5	1
Deschutes	258	38.1	11	3	63.3	5	84.5	2
Truckee	99	32.6	1	0	54.5	1	75.2	1
Methow Valley	77	6.9	8	1	13.4	4	18.8	2
Spokane	74	12.4	4	2	22.5	1	32.5	1
Boise	72	5.3	18	7	9.2	11	13.8	5
Missoula	58	10.6	5	1	17.3	1	26.9	1
Calgary	40	5.2	10	7	10.6	4	15.6	2
Helena	25	17.6	1	0	20.4	1	24.0	1
Duluth	18	2.3	4	2	4.1	3	6.3	3

Mean Neighbors = average number of other sensors within the given radius. Components = number of connected components at that radius. Isolated = sensors with zero neighbors at that radius. Sorted by total device count.

Table 4: Smoke Event Characteristics

Site	Source	Dates	Duration (days)	Fire Det. (<50 km)	Fire Det. (<200 km)	Dominant Fire Dir.	Smoke Days	Mean Smoke Coverage (%)
Boise	WF	Aug 28 – Sep 12, 2024	16	164	60,728	NE	15	93.0
Calgary	WF	Jul 20 – Jul 31, 2024	12	0	10,091	SW	12	100.0
Deschutes	Rx	May 14 – May 17, 2024	4	137	616	S	2	16.2
Deschutes	WF	Sep 3 – Sep 15, 2024	13	4,321	65,688	E	12	85.1
Duluth	WF	Aug 11 – Aug 16, 2024	6	0	5	SW	6	100.0
Helena	WF	Jul 20 – Jul 31, 2024	12	5	608	W	12	100.0
Los Angeles	WF	Jan 6 – Jan 12, 2025	7	12,315	12,427	NW	6	68.7
Methow Valley	WF	Aug 10 – Aug 21, 2024	12	503	1,802	W	10	83.3
Missoula	WF	Aug 31 – Sep 12, 2024	13	1,143	24,484	SW	13	100.0
Spokane	Rx	Nov 5 – Nov 12, 2024	8	235	3,957	SE	0	0.0
Truckee	Rx	Nov 7 – Nov 10, 2024	4	11	944	W	1	9.0

Fire detections from NOAA Hazard Mapping System (HMS) satellite fire point data, cumulated across all event days. Smoke days = days with any HMS smoke plume intersecting the zone. Smoke coverage = percentage of zone area covered by HMS smoke polygons, averaged across all event days.

Individual Site Characterizations

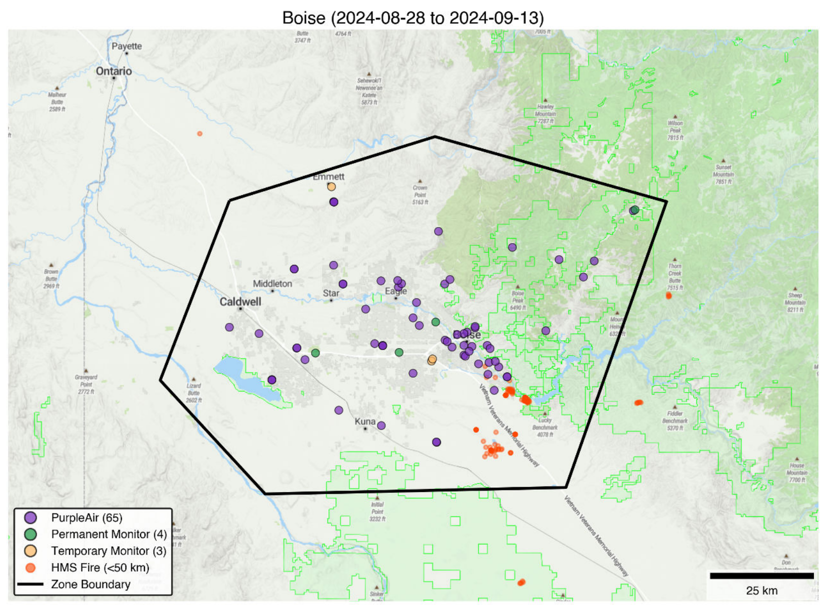
<u>Boise, Idaho</u>

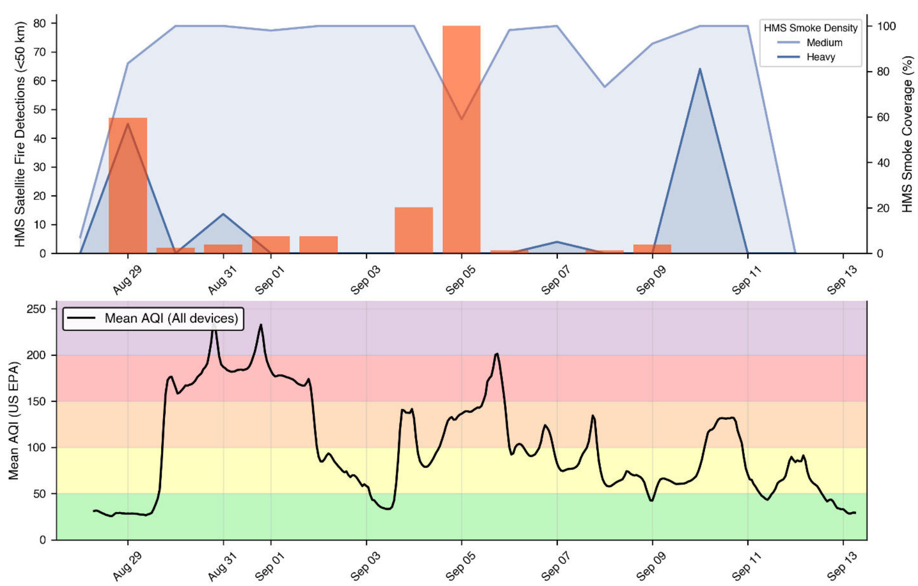
Setting: 4,046 km² of high desert steppe and irrigated agricultural land with the forest along the north/eastern boundary. Idaho, United States.

Terrain: Mean elevation 998 m with 1,634 m of relief. Moderate slopes overall (8.2°), but a flat valley floor transitioning to rolling foothills in the east. Dominant land cover is grassland/herbaceous (25%) with significant cultivated cropland (20%) and shrub/scrub (21%).

Sensor inventory: 72 devices in the training dataset (17.8 per 1,000 km²)

Smoke event (Aug 28 – Sep 12, 2024): 16-day smoke event during peak fire season. Only 164 HMS fire detections within 50 km but 60,728 within 200 km (dominant direction NE), indicating regional smoke transport from large fire complexes across central Idaho and eastern Oregon rather than direct local fire impact. HMS smoke plumes covered the zone on 15 of 16 days (mean 93.0% coverage).





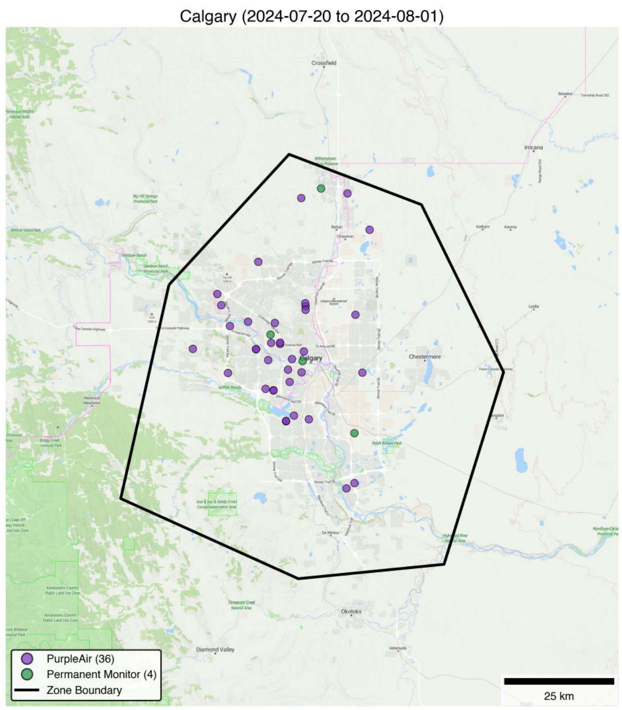
Calgary, Alberta

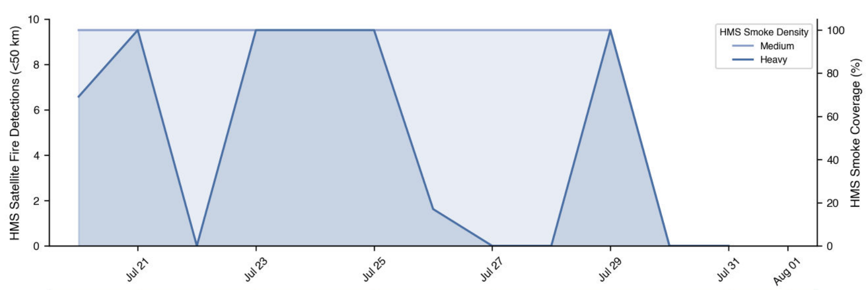
Setting: 2,321 km² urban/suburban area in a predominantly flat prairie with the Rocky Mountain foothills to the west. Alberta, Canada.

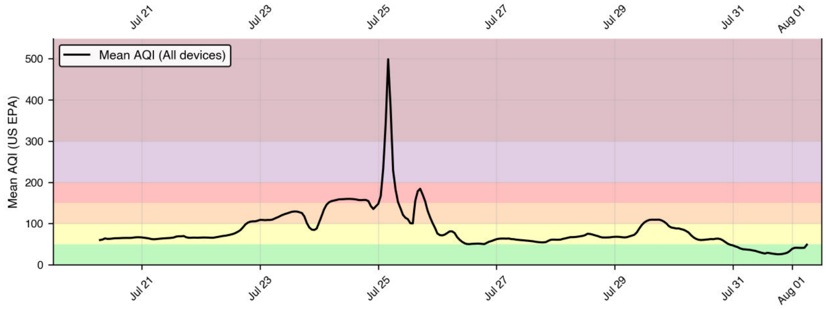
Terrain: Mean elevation 1,103 m with only 472 m of relief and very low mean slope (2.9°).

Sensor inventory: 40 devices in the training dataset (17.2 per 1,000 km²)

Smoke event (Jul 20 – Jul 31, 2024): 12-day smoke event. Zero fire detections within 50 km and only 16 within 100 km, but 10,091 within 200 km (dominant direction SW), indicating entirely transported regional smoke from British Columbia and other wildfires. Despite the absence of nearby fire activity, HMS smoke plumes covered 100% of the zone on all 12 event days.







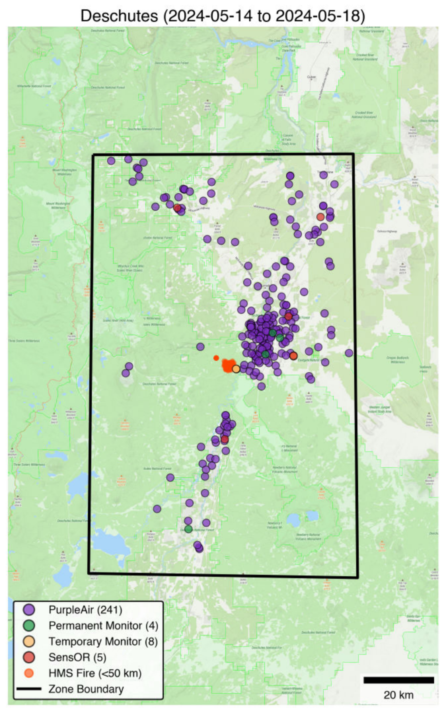
Deschutes County, Oregon

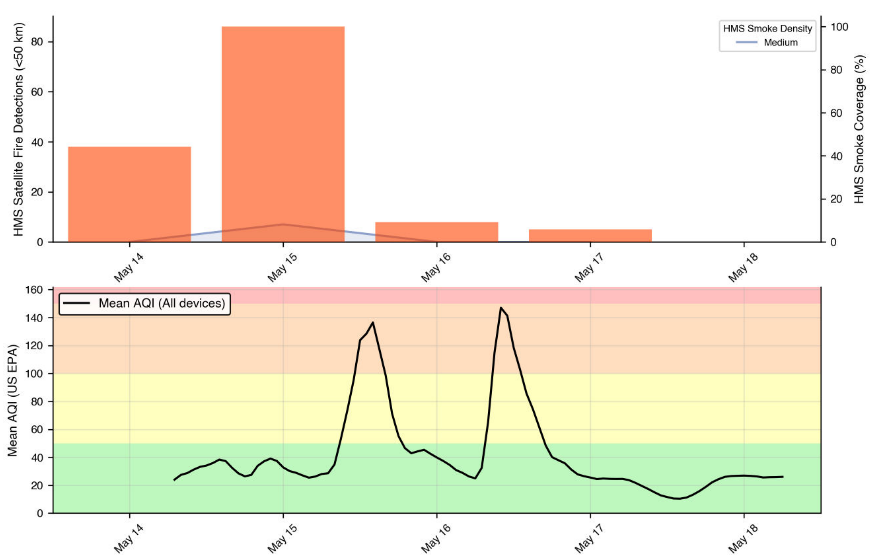
Setting: 4,618 km² spanning the east slope of the Oregon Cascades, centered on the Bend–Redmond–La Pine corridor. Oregon, United States.

Terrain: Mean elevation 1,357 m with 2,422 m of relief. Moderate mean slope (4.9°).

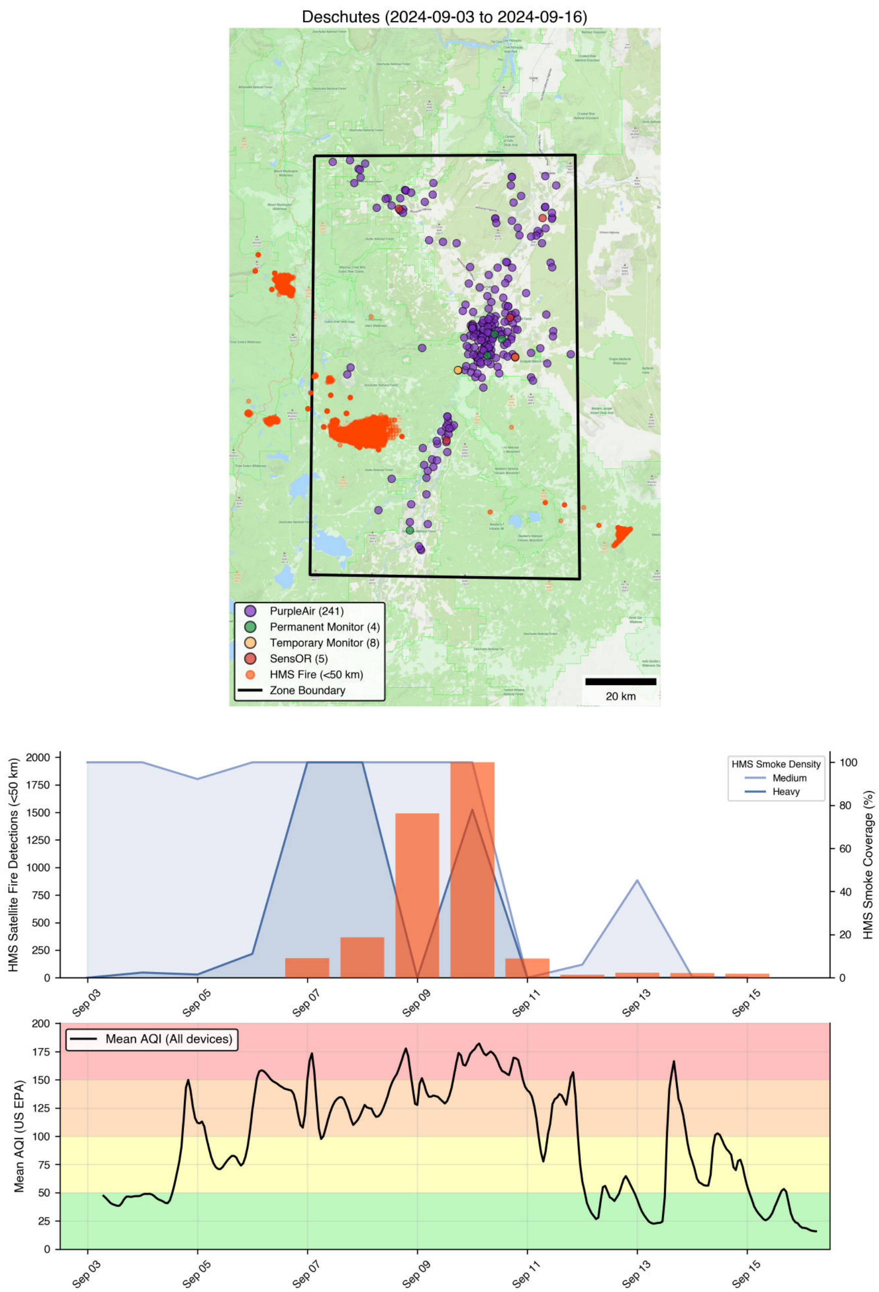
Sensor inventory: 258 devices in the training dataset (55.8 per 1,000 km²)

Prescribed fire smoke event (May 14 – May 17, 2024): 137 fire detections within 50 km (dominant direction E) from planned burns in the Deschutes National Forest. Only 2 of 4 days had any HMS smoke coverage (mean 16.2%), and no Heavy smoke was detected.





Wildfire smoke event (Sep 3 – Sep 15, 2024): 13-day smoke event at the peak of fire season. 4,321 fire detections within 50 km and 65,688 within 200 km, from multiple large fire complexes across Oregon. Smoke covered the zone on 12 of 13 days (mean 85.1%).


Deschutes (2024-09-03 to 2024-09-16)
PurpleAir (241)
Permanent Monitor (4)
Temporary Monitor (8)
SensOR (5)
HMS Fire (<50 km)
Zone Boundary
20 km
HMS Satellite Fire Detections (<50 km)
HMS Smoke Coverage (%)
HMS Smoke Density
Medium
Heavy
Sep 03
Sep 05
Sep 07
Sep 09
Sep 11
Sep 13
Sep 15
Mean AQI (All devices)
Mean AQI (US EPA)

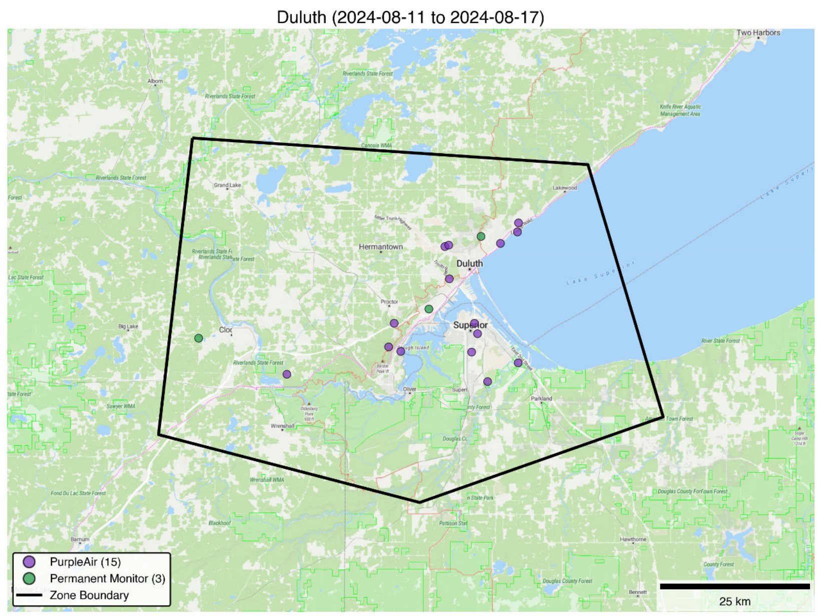
Duluth, Minnesota

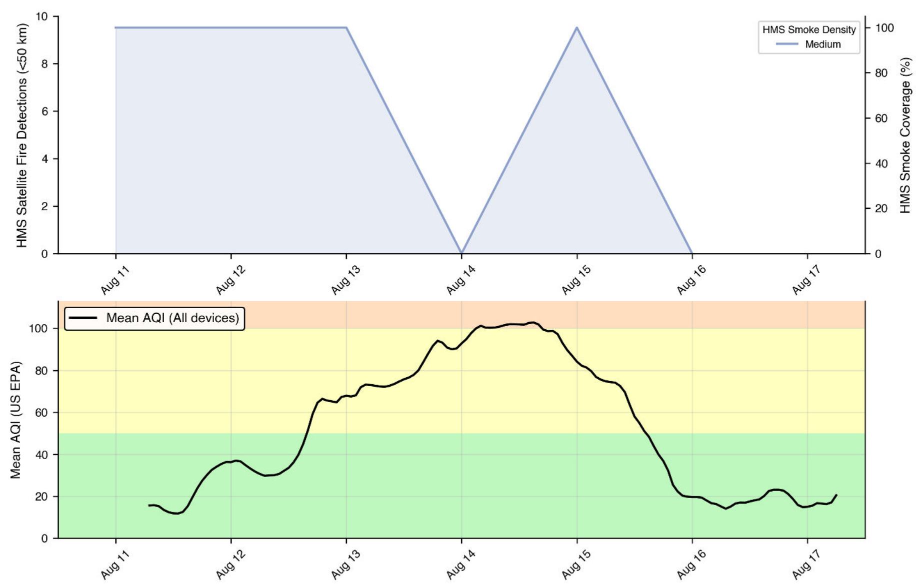
Setting: 1,860 km² on the western end of Lake Superior. Our only site east of the Rocky Mountains. Minnesota, United States.

Terrain: Mean elevation 305 m with only 285 m of relief. The lowest-elevation and flattest terrain in the study (mean slope 2.2°).

Sensor inventory: 18 devices in the training dataset (9.7 per 1,000 km²)

Smoke event (Aug 11 – Aug 16, 2024): 6-day smoke event. Zero fire detections within 50 km and only 5 within 200 km, yet HMS smoke plumes covered 100% of the zone on all 6 event days. This represents purely transported regional smoke, likely from Canadian wildfires.





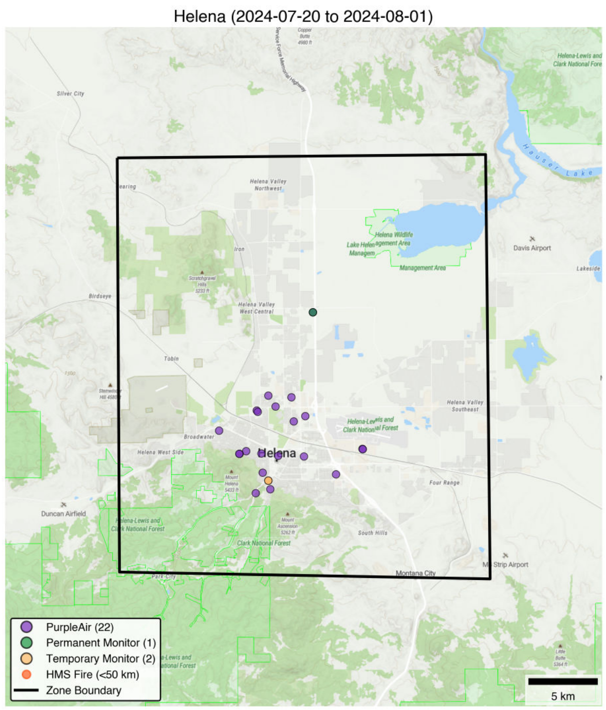
<u>Helena, Montana</u>

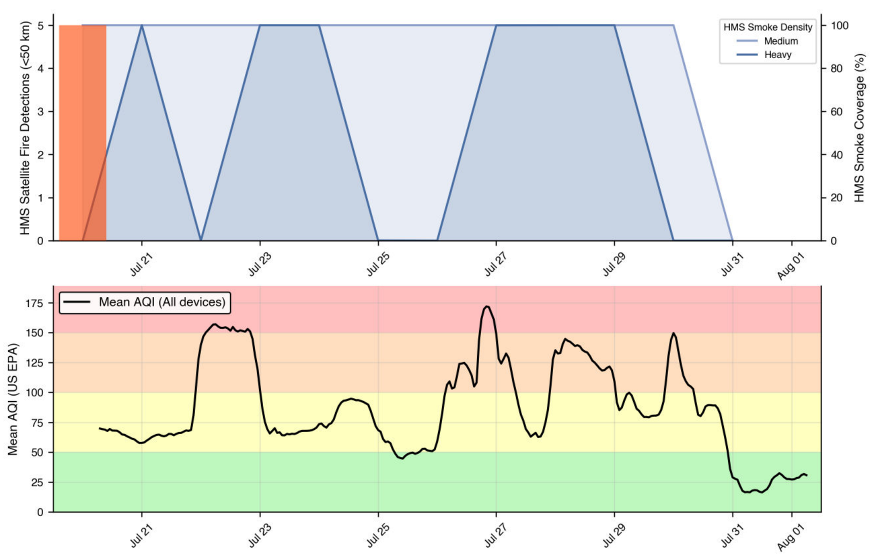
Setting: 378 km² situated in a narrow mountain valley, Montana, United States.

Terrain: Mean elevation 1,243 m with 693 m of relief and moderate mean slope (5.3°).

Sensor inventory: 25 devices in the training dataset (66.0 per 1,000 km²)

Smoke event (Jul 20 – Jul 31, 2024): 12-day smoke event. Only 5 HMS fire detections within 50 km but 608 within 200 km (dominant direction W), indicating regional smoke transport from fires to the west. HMS smoke plumes covered 100% of the zone on all 12 event days.





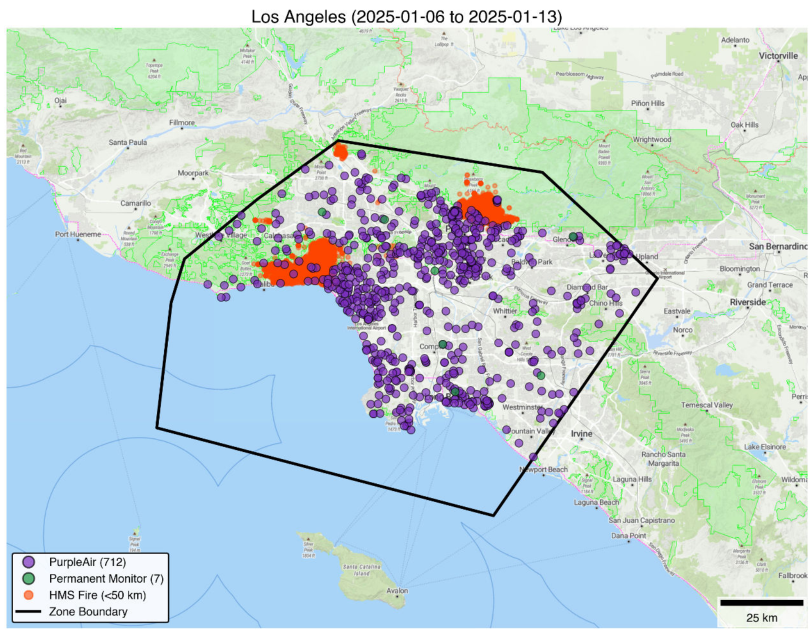
Los Angeles, California

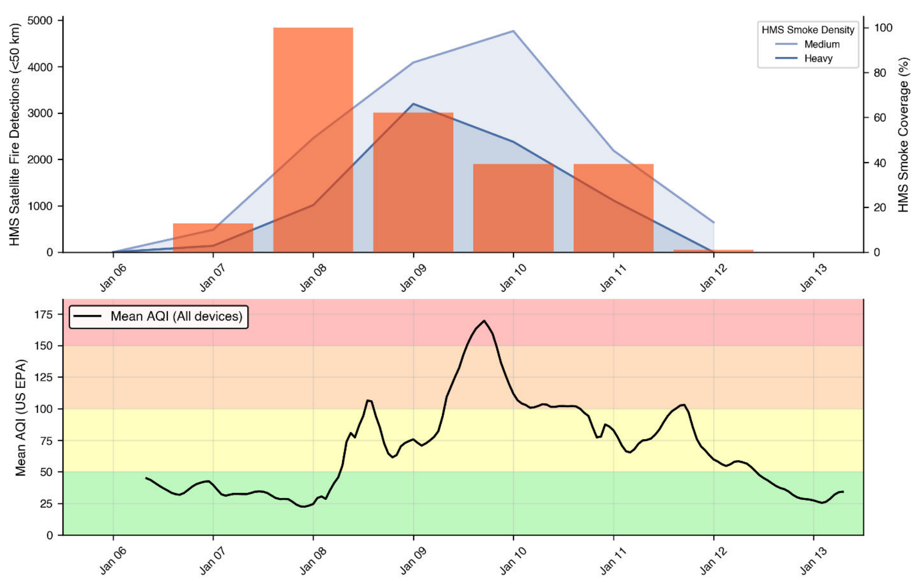
Setting: 7,960 km² spanning the Los Angeles Basin from the Pacific coast to the San Gabriel and San Bernardino Mountains. California, United States.

Terrain: Mean elevation 173 m, with 1,882 m of relief. Mean slope 5.6°.

Sensor inventory: 719 devices in the training dataset (90.4 per 1,000 km²)

Smoke event (Jan 6 – Jan 12, 2025): 7-day smoke event. The January 2025 Palisades and Eaton fires produced 12,315 fire detections within 50 km (nearly all within the zone itself). Smoke covered the zone on 6 of 7 days (mean 68.7%), with substantial Medium and Heavy smoke density.





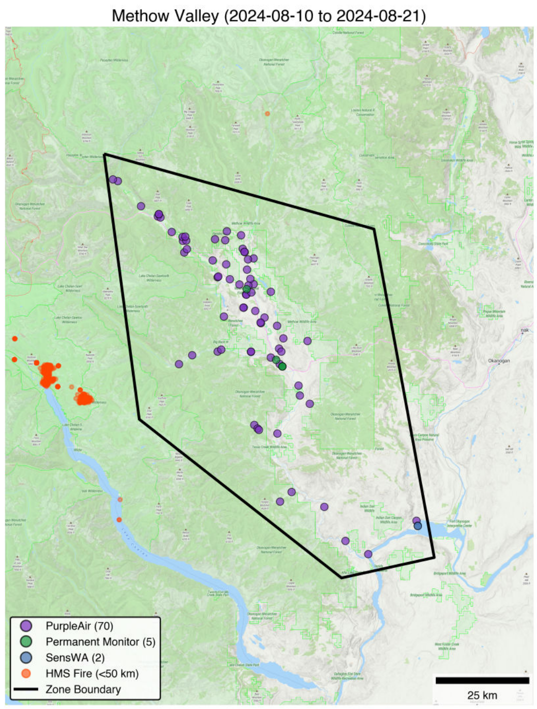
Methow Valley, Washington

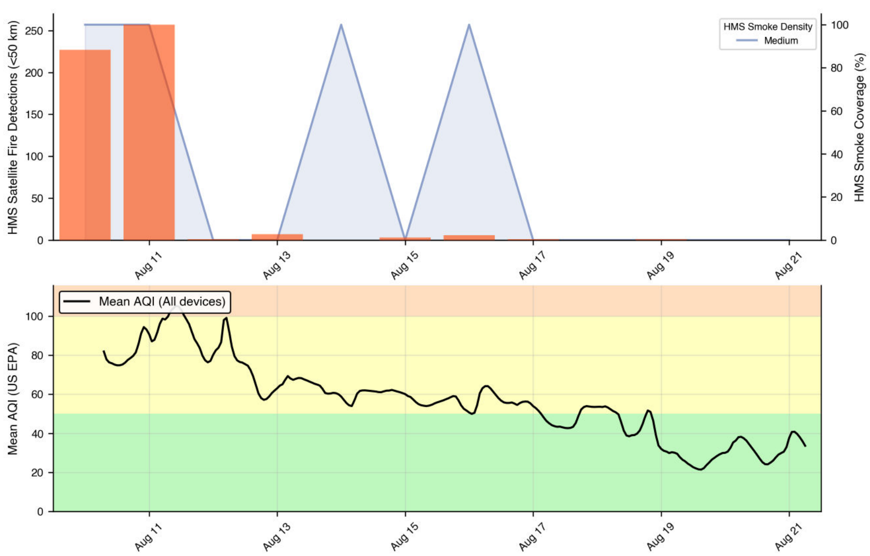
Setting: 2,781 km² mountain valley in the North Cascades, spanning from the Cascade crest to the Okanogan Highlands in Washington, United States.

Terrain: Mean elevation 1,071 m with 2,475 m of relief. Mean slope of 18.2°.

Sensor inventory: 77 devices in the training dataset (27.6 per 1,000 km²)

Smoke event (Aug 10 – Aug 21, 2024): 12-day smoke event. 503 fire detections within 50 km (dominant direction W), driven by fires on the west side of the Cascade crest. Within 200 km, 1,802 total detections. Smoke covered the zone on 10 of 12 days (mean 83.3%).





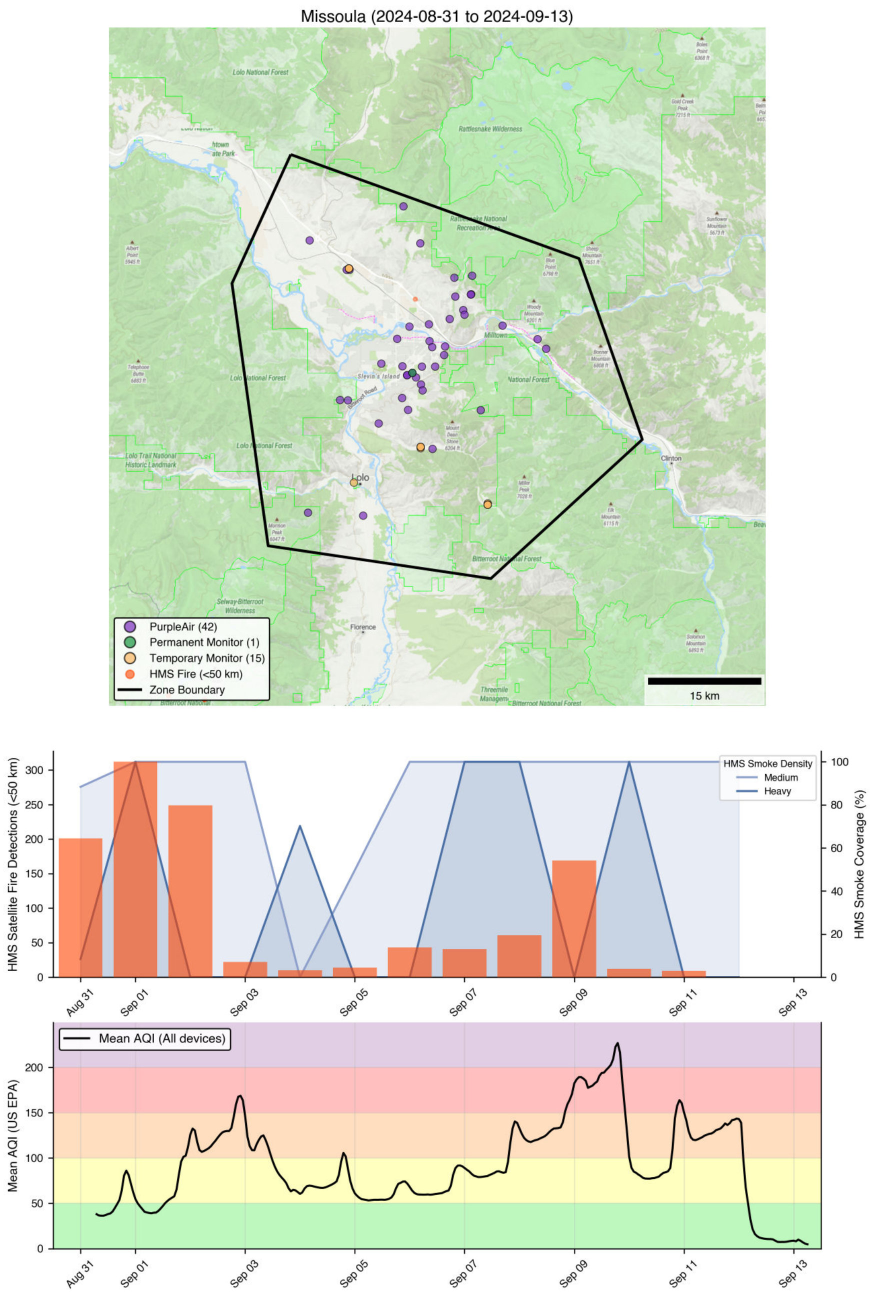
Missoula, Montana

Setting: 1,003 km² encompassing the city of Missoula and surrounding mountains in western Montana, United States.

Terrain: Mean elevation 1,269 m with 1,225 m of relief. Mean slope of 15.7°.

Sensor inventory: 58 devices in the training dataset (57.7 per 1,000 km²)

Smoke event (Aug 31 – Sep 12, 2024): 13-day smoke event. 1,143 fire detections within 50 km (dominant direction S) and 24,484 within 200 km (SW), indicating both nearby smoke potential and regional smoke potential. HMS smoke plumes covered 100% of the zone on all 13 event days.



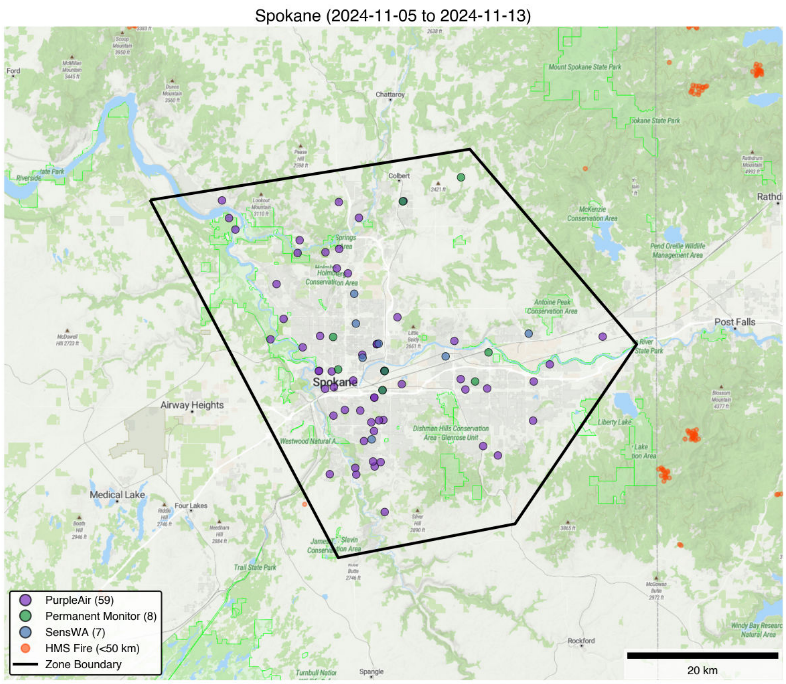
<u>Spokane, Washington</u>

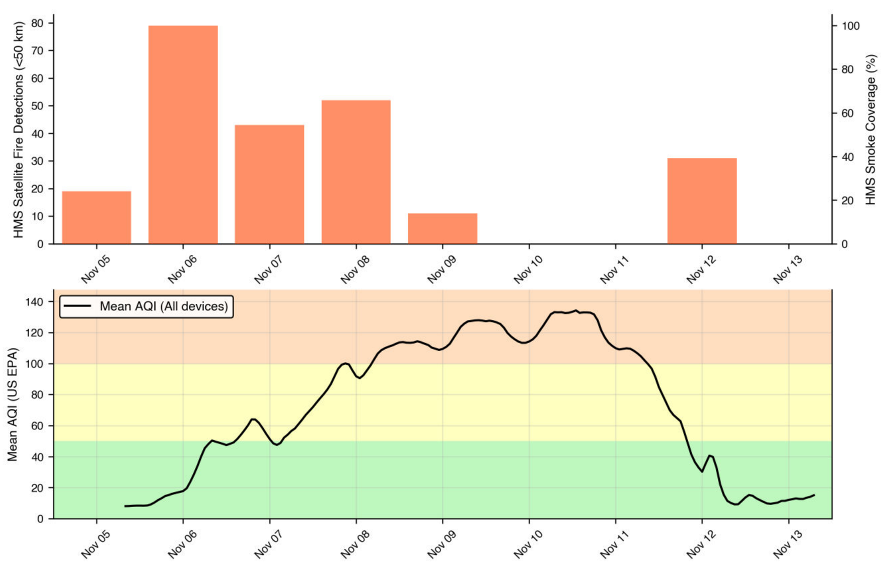
Setting: 971 km² covering Spokane and surrounding shrubland/forest mosaic in Eastern Washington, United States.

Terrain: Mean elevation 654 m with 653 m of relief. Moderate mean slope (5.7°).

Sensor inventory: 74 devices in the training dataset (76.0 per 1,000 km²)

Smoke event (Nov 5 – Nov 12, 2024): 8-day fall prescribed fire smoke event. 235 fire detections within 50 km (dominant direction NE) and 3,957 within 200 km (SE). No HMS smoke plumes intersected the zone on any event day.





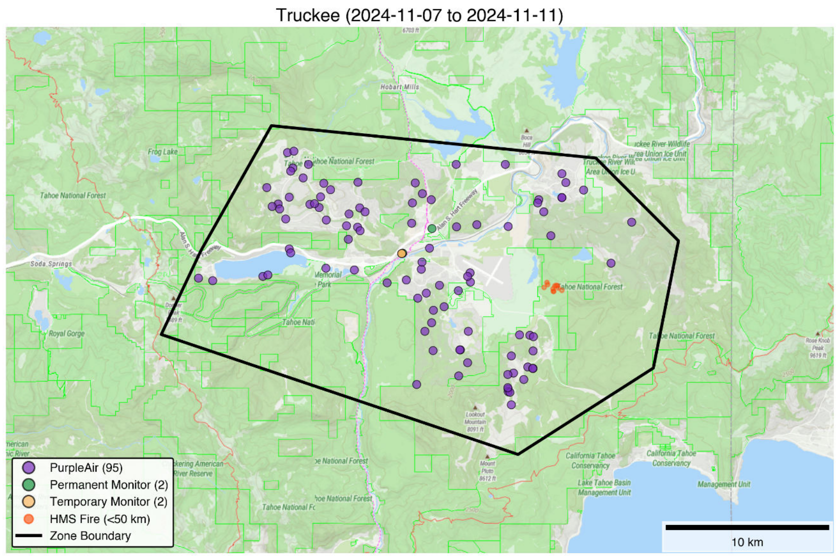
Truckee, California

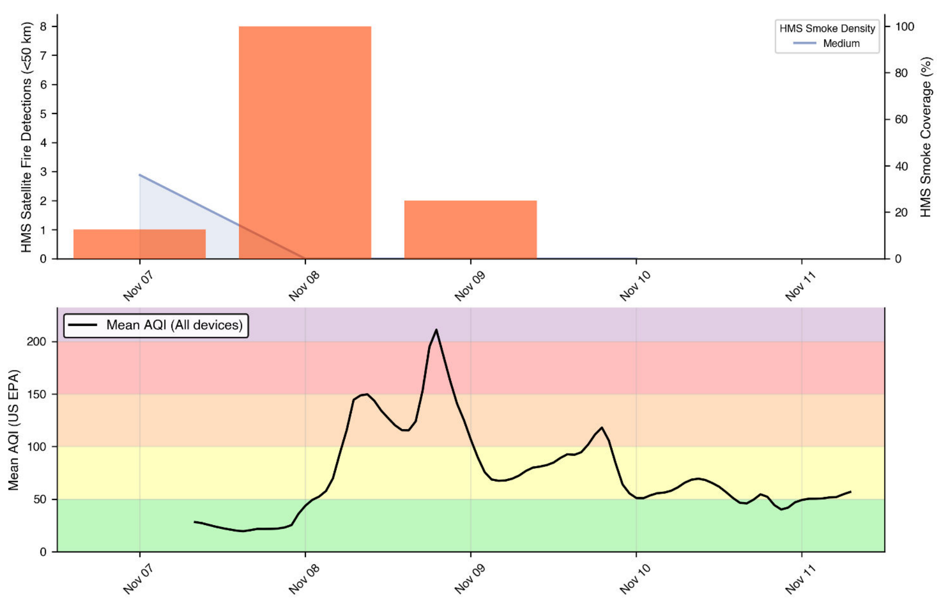
Setting: 260 km² in the northern Sierra Nevada, adjacent to Lake Tahoe, California, United States.

Terrain: Mean elevation 1,954 m. Relief of 876 m and moderate mean slope (9.5°).

Sensor inventory: 99 devices in the training dataset (379.9 per 1,000 km²)

Smoke event (Nov 7 – Nov 10, 2024): 4-day late-season prescribed fire smoke event. Only 11 fire detections within 50 km and 944 within 200 km (dominant direction W), indicating potential smoke transport from up-drainage burns near Lake Tahoe.





Data Sources

- **Sensor locations and classifications:** PurpleAir, EPA AirNow, Oregon DEQ, Washington Ecology. Units described in Methods.
- **Terrain and elevation:** USGS 3D Elevation Program (3DEP) 30 m DEM for U.S. sites; SRTM 30 m for Calgary. Processed via Google Earth Engine.
- **Land cover:** NLCD 2021 (U.S. sites) via Google Earth Engine. Not available for Calgary (Canadian territory).
- **Fire detections:** NOAA Hazard Mapping System (HMS) satellite fire point data. Daily fire detection shapefiles.
- **Smoke plumes:** NOAA HMS smoke polygon shapefiles with Light/Medium/Heavy density classification. Area coverage computed by intersecting smoke polygons with zone boundaries in Albers Equal Area (ESRI:102008) projection, with per-density dissolve to avoid double-counting overlapping polygons.
- **Zone boundaries:** Convex hulls of sensor locations per zone, buffered to encompass surrounding terrain context. Stored as `event_sites.geojson` (EPSG:4326).
- **Basemap:** © MapTiler; map data © OpenStreetMap, distributed under the Open Data Commons Open Database License (ODbL) v1.0; for further information see openstreetmap.org/copyright